

Index

- (A, B) -invariant subspace, 207
- A -invariant mod B , 207
- absolutely continuous, 471
- accessibility, 141, 157
 - discrete-time, 181
 - Lie algebra, 154
 - rank condition, 154
- achieved Bellman function, 355
- Ackermann's formula, 188
- action, 29
- A/D conversion, 24
- adaptive control, 24
- adjoint, 276
- admissible, 44
 - feedback, 360
 - pair, 27
- airplane example, 90, 112, 117, 188, 272, 303, 325, 330
- algebraic Riccati equation, 381
 - dual, 390
- algebraic set, 97
- almost everywhere convergence, 468
- analog to digital, 24
- analytic
 - continuous-time system, 268
 - function, 484
- analytically varying system, 113
- ARC, 154
- ARE, 381
- ARMA, 312
- assignable polynomial, 185
- asycontrollable, 211
- asymptotic
 - controllability, 211
 - observability, 317, 329
 - stability, 211
- attractivity, 258
- attractor, 212
- automata, 33, 493
- autonomous system, 29
- autoregressive moving average, 312
- backstepping, 242
- Banach space, 308, 463
- bang-bang principle, 436
- behavior, 46
 - continuous-time, 46
 - discrete-time, 32
 - in Willems' sense, 77
 - integral, 49
- Bellman function, 355, 392, 443
- Bellman-Gronwall Lemma, 474
- bilinear
 - approximations, 79
 - system, 74
- Bode plot, 334
- bounded-input bounded-output, 327
- Brockett's
 - example, 162, 164, 253, 408, 417
 - Stabilization Theorem, 253
- broom balancing, 91, 188, 272, 303, 326, 330
- Brunovsky form, 191
- calculus of variations, 409
- canonical
 - realization, 284, 286
 - system, 284, 305
 - triple, 284

- causality, 78
 - strict, 31
- Cayley-Hamilton Theorem, 88
- Chaetaev's Theorem, 237, 259
- chaos in feedback systems, 258, 313
- $\mathcal{C}^k$
 - continuous-time system, 45
 - discrete-time system, 39
- classical
 - design, 15
 - dynamical system, 29
- Closed Graph Theorem, 308
- closed-loop, 4
 - system, 212
- coercive mapping, 220
- complete
 - controllability, 83
 - i/o behavior, 32
 - system, 29
- complex differential equations, 483
- computational complexity, 181
- concatenation, 26
- conjugate partition, 193
- constant-rank distribution, 169
- continuous dependence
 - of eigenvalues, 456
 - of real parts of zeros, 457
 - of singular values, 458
 - of zeros, 456
- continuous-time, 11
 - behavior, 46
 - linear behavior, 48
 - linear system, 47
 - system, 44
 - with outputs, 45
- control, 3, 27
 - bounded, 117
 - infinite-length, 28
 - positive, 139
 - value, 27, 30
- control-affine system, 57, 154, 246, 281, 362, 392, 406
- control-Lyapunov function, 219, 224, 247
- controllability, 81, 83
 - asymptotic, 211
 - complete, 83
 - decomposition, 93
 - discrete-time nonlinear, 181
 - form, 184, 292
 - genericity of, 96
 - Gramian, 109
 - indices, 190, 193
 - local, 124
 - matrix, 92
 - output, 112
 - sampled, 100
 - structural, 138
 - to the origin, 84
 - weak, 141
- controllable, 13
 - and observable triples, 283
 - modes, 94
 - pair of matrices, 92
 - polynomial, 94
 - space, 92
- controller form, 184, 187
- convergent
 - matrix, 212, 492
 - polynomial, 216
- convolution, 50
 - discrete, 39
- convolutional codes, 78
- coprime, 338
 - fractional representation, 338
- cost
 - function, 347
 - instantaneous, 350
 - terminal, 350
 - total, 350
 - trajectory, 349
- cybernetics, 23
- cyclic matrix, 455
- damping
 - control, 240, 394
 - factor, 15
- dashpot, 6
- dc gain, 334
- deadbeat

- control, 9
 - observer, 320
- decentralized control, 77
- delay-differential, 79
- Delchamp's Lemma, 389
- descriptor systems, 77, 256
- detectability, 329
 - nonlinear, *see* OSS
- detectable, 317
- differentiable
 - continuous-time system, 45
 - discrete-time system, 39
 - Fréchet, 463
 - Gateaux, 464
 - mapping, 461
 - system, 39, 45
- differential, 462, 463
 - algebra, 312
 - inclusions, 77
 - inequalities, 229
- dimension, 36, 47
- directional derivative, 463
- directional derivative along vector
 - field, 142, 218
- discrete-event systems, 76
- discrete-time
 - Hurwitz matrix, 492
 - i/o behavior, 32
 - linear behavior, 37
 - linear system, 36
 - system, 32
- distance to uncontrollable systems, 98
- distinguishability, 262
 - final-state, 266
 - instantaneous, 262
- distributed systems, 79
- distribution, 169
- disturbance, 371
 - decoupling, 20, 207
 - rejection, 20, 207
- domain of attraction, 250
- duality, 20
 - control and observation, 261, 274
- dynamic
 - feedback, 18, 323
 - stabilization, 323
- dynamic programming
 - equation, 359
 - forward, 355
- econometrics, 312
- eigenspace, 452
- empty sequence, 26
- equibounded, 470
- equilibrium
 - pair, 40, 51, 317
 - state, 40, 51
- escape time, 480
- essentially bounded, 470
- Euler-Lagrange equation, 411
- event, 81
- evolution equation, 33
- exponential
 - growth, 332
 - stability, 259
- families of systems, 296
- feedback, 9, 183
 - admissible, 360
 - dynamic, 323
 - equivalence, 190, 197
 - gain, 4
 - integral, 18, 22
 - linearizable, 198
 - linearization, 21, 257
 - piecewise analytic, 260
 - piecewise linear, 260
 - proportional, 4
 - stabilization, 213, 236, 240, 384, 392
 - universal formula, 246
- fibre bundle, 295
- final-state observable system, 266
- finite dimensional system, 36
- finite fields, 76, 78
- Fliess series, 80
- flyballs, 22

- forward dynamic programming, 355
- Fréchet differentiable, 463
- free group action, 287
- frequency
 - design, 15
 - natural, 15
 - of system, 102
- Frobenius' Theorem, 175
- full rank map, 464
- function
 - of class $\mathcal{C}^1$, 463
 - of class $\mathcal{C}^\omega$, 484
- fundamental solution, 488
- gain
 - at frequency ω , 334
 - margin, 388, 393
 - nonlinear, 331
 - scheduling, 5
- general systems theory, 76
- generalized inverse, 450
- generating series, 80, 298
- genericity of controllability, 96
- geometric
 - linear theory, 207, 218
 - multiplicity, 452
- gradient descent, 415
- Gramian
 - controllability, 109
 - observability, 279
- Grassman manifold, 386
- Gronwall Lemma, 474
- Hamilton-Jacobi-Bellman equation, 358, 391
- Hamiltonian, 360, 403
 - control systems, 313
 - matrix, 369
- Hankel
 - approximation, 311
 - matrix, 285, 310
 - for automata, 310
 - infinite, 288
- Hardy space, 338
- Hautus
 - detectability condition, 318
 - Lemma, 94
 - observability condition, 272
 - rank condition, 94
- hereditary, 79
- Heymann's Lemma, 256
- H_∞ optimization, 396
- Hilbert Basis Theorem, 273
- HJB, 358, 391
- Hurwitz, 492
 - matrix, 212, 492
 - polynomial, 216
- hybrid systems, 24, 76
- identification, 20, 312
 - techniques, 24
- identity axiom, 26
- Implicit Function Theorem, 465
- impulse response, 38, 50
- infinite
 - dimensional systems, 79
 - Hankel matrix, 288
 - length controls, 28
- initial state, 28, 31
- initialized system, 31
- input, 3, 27
 - value, 27, 30
- input-to-state stability, 330
- input/output
 - pairs, 300
 - stability, 10
- Instability Theorem, 237
- instantaneous distinguishability, 262
- integral
 - behavior, 49
 - feedback, 18
- integrator
 - backstepping, 242
 - double, 228
 - triple, 259
- interconnection
 - well-posed, 322
- internally stable, 328

- invariance principle, 226
- invariant distribution, 169, 258
- inverted pendulum, 91, 188, 272, 303, 326, 330
- involutive distribution, 169
- i/o
 - behavior, 20, 30
 - complete, 32
 - of system, 31
 - time-invariant, 32
 - equation, 301
 - map, 31
- i/s behavior of system, 31
- ISS, 330, 345
- Jacobian, 462
- Jordan
 - block, 452
 - form, 452
- $\mathcal{K}$ -function, 330
- Kalman
 - controllability decomposition, 93
 - filtering, 18, 321, 376
 - steady-state, 390
 - rank condition
 - for controllability, 89
 - for observability, 271
- $\mathcal{K}_\infty$ -function, 330
- $\mathcal{KL}$ -function, 330
- knife edge, 164
- Kronecker indices, 190
- Lagrangian subspace, 386
- Laplace transform, 332
- LaSalle invariance principle, 226
- Laurent
 - expansion, 300
 - series, 297
- least-squares, 450
 - recursive, 395
- length of control, 27
- Lie
 - algebra, 143
 - generated, 143
- bracket, 142
- derivative, 142, 218
- linear
 - behavior
 - continuous-time, 48
 - discrete-time, 37
 - system, 11, 47, 82, 264
 - continuous-time, 47
 - discrete-time, 36
 - finite-dimensional, 36, 47
 - with output, 17
- linear growth on controls, 65
- linear-quadratic problem, 364, 369
- linearization, 207
 - feedback, 198
 - of continuous-time system, 51
 - of discrete-time system, 39
 - under feedback, 21
- linearization principle, 5
 - for controllability, 125
 - for observability, 280
 - for stability, 234
- local in time, 45
- locally
 - absolutely continuous, 471
 - controllable, 124
 - integrable, 470
 - observable, 280
- LQG, 395
- Lyapunov function, 219, 224, 391
 - converse theorems, 258
- machine, 26
- Markov
 - parameters, 38, 284
 - sequence, 284
- Massera's Theorem, 258, 260
- matrix
 - identities over rings, 460
 - norm, 447
- maximal solution, 477
- Maximum Principle, 418, 433
 - time-optimal, 434
- McMillan degree, 299

- Mean Value Theorem, 60
- measurable function, 468
- measurement, 16
 - map, 27
 - value, 27, 30
- memoryless i/o map, 78
- minimal
 - realization, 286
 - surface of revolution, 413
 - system, 308
 - triple, 286
- Minimum Principle, 418
- moduli problems, 296
- Moore-Penrose inverse, 450
- multidimensional systems, 77
- multipliers, 348
- natural frequency, 15
- Nerode equivalence relation, 309
- neural networks, 129, 139, 312
 - controllability, 130
 - observability, 310
- Newton's method, 416
- next-state map, 33
- no-drift systems, 158
- noise, effect of, 17
- nonhomogeneous linear systems, 96
- nonlinear gain, 331
- nontriviality axiom, 26
- normal problem in optimal control, 437
- normed space, 463
- NP-hard, 181
- null-controllability, 84, 99
- numerical
 - methods for optimal control, 415
 - pole-shifting, 188
- Nyquist criterion, 342
- observability, 20, 262, 263
 - asymptotic, 317, 329
 - decomposition, 272
 - form, 292
 - Gramian, 279
 - rank condition, 282
 - sampled, 275
- observable
 - locally, 280
 - pair, 272
 - reduction, 305
 - system, 263
- observables, 311
- observation space, 282
- observer, 4, 18, 318
 - optimal, 378
 - reduced order, 320
- operational amplifiers, 6
- operator norm, 447
- optimal
 - control data, 350
 - cost-to-go, 355
 - observer, 378
 - path, 355, 392
 - trajectory, 355, 392
- OSS, 331
- output
 - controllability, 112
 - feedback, 17
 - stabilization, 218
 - value, 27, 30
- output-to-state stability, 331
- Padé approximation, 311
- pair
 - equilibrium, 40, 51
 - of matrices, 92
- parity-checker, 34, 309
- partial
 - derivative, 465
 - realization, 311
- partition of integer, 191
- path, 28
- path planning, 181
- PD, 4
- Peano-Baker formula, 489
- pencil of matrices, 190
- pendulum, inverted, 91, 188, 272, 303, 326, 330
- performance index, 347

- phase
 - lag, 334
 - lead, 334
 - shift, 334
- PID, 19
- piecewise
 - analytic feedback, 260
 - constant control, 63
 - continuous functions, 42
 - linear feedback, 260
- pointwise in distribution, 169
- Pole-Shifting Theorem, 11, 14, 186
- polynomial system, 273, 274
- Pontryagin's Maximum (or Minimum) Principle, 348
- positive definite function, 219
- proper, 336
 - function, 219
 - group action, 295
- proportional feedback, 4
- proportional-derivative, 4
- proportional-integral-derivative, 19
- proximal subgradients, 394
- pseudoinverse, 107, 450
- pulse-width modulated control, 260
- quantization, 24
- radially unbounded, 220
- rank
 - of Markov sequence, 287, 312
- Rank Theorem, 464
- rational, 298
- reachability, 81
 - matrix, 92
- reachable
 - set, 84
 - space, 92
- readout map, 27
- realizable, 284
- realization, 20, 32, 284
 - canonical, 284, 286
 - in controllability form, 292
 - in observability form, 292
 - minimal, 286
 - partial, 311
- recurrent nets, 129, 139
 - controllability, 130
 - observability, 310
- recursive Markov sequence, 290
- reduced order observers, 320
- reference signal, 15, 368, 371
- regular synthesis, 394
- relative degree, 205
- response map, 30
- restriction axiom, 26, 30
- reversible systems, 84
- rhs, 43
- Riccati Differential Equation, 364
 - for filtering, 378
 - indefinite, 395
- right-hand side, 43
- rigid body, momentum control, 163, 241, 245
- robust control, 23
- sample-and-hold, 7
- sampled
 - controllable, 100
 - observable, 275
 - system, 72
- sampling, 138
 - times, 7
- Sampling Theorem, 310
- satellite, 163, 241, 245
- Schur matrix, 492
- selection theorems, 78
- semigroup axiom, 26
- semirings, 493
- sequential machines, 33
- set-valued map, 77
- Shannon's Theorem, 138, 310
- shift operator, 289
- shopping cart, 164
- similar systems, 183, 286
- similarity of realizations, 286
- singular
 - systems, 77, 256
 - value decomposition, 449
 - values, 448

- slice, 146
- sliding mode control, 260
- small-time local controllability, 139, 432
- smooth
 - continuous-time system, 45
 - discrete-time system, 39
 - mapping, 461
 - solution of optimal control problem, 359
 - stabilizability, 236
 - vector field, 141
- smoothly varying system, 113
- solution of ode, 474
- space
 - control-value, 26
 - input-value, 26
 - measurement-value, 27
 - output-value, 27
- Spectral Mapping Theorem, 454
- stability, 258
 - asymptotic, 211
 - exponential, 259
 - input-to-state, 330, 345
 - internal, 328
 - non-asymptotic, 237
 - of time-varying systems, 232
 - output-to-state, 331
- stabilizability, 236
- stabilizable, 214
- state, 12, 27
 - equilibrium, 40, 51
 - initial, 28, 31
 - space, 11, 26
 - terminal, 28
- state-output, 77
- static output feedback, 17
- steady-state
 - error, 19
 - Kalman filter, 390
 - Kalman gain, 390
 - output, 334
- steepest descent, 417
- STLC, 432
- stochastic
 - filtering, 18, 376
 - state estimation, 18, 376
 - systems, 23
- strict causality, 31
- strictly proper, 336
- strongly locally controllable, 129
- structural controllability, 138
- submanifold, 146
- superposition principle, 264
- Sylvester equation, 230
- system, 26, 77
 - affine, 96
 - affine in controls, 57, 154, 246, 281, 362, 392, 406
 - analytically varying, 113
 - asymptotically observable, 317
 - autonomous, 29
 - bilinear, 74
 - canonical, 284, 305
 - chaotic, 313
 - classical dynamical, 29
 - complete, 29
 - continuous-time, 44
 - analytic, 268
 - smooth, 45
 - delay-differential, 79
 - descriptor, 77, 256
 - detectable, 317
 - differentiable, 39, 45
 - discrete-event, 76
 - discrete-time, 32
 - differentiable, 39
 - smooth, 39
 - distributed, 79
 - final-state observable, 266
 - Hamiltonian, 313
 - hereditary, 79
 - hybrid, 24, 76
 - infinite dimensional, 79, 494
 - initialized, 31
 - input-to-state stable, 330, 345
 - internally stable, 328
 - isomorphism, 306
 - ISS, 330, 345
 - large scale, 77

- linear, 47, 82, 264, 493
 - nonhomogeneous, 96
- minimal, 308
- morphism, 306
- multidimensional, 77
- observable, 263
- of class $\mathcal{C}^k$, 45
- on group, 313
- OSS, 331
- output-to-state stable, 331
- over ring, 79
- piecewise linear, 138
- polynomial, 273, 313
- reversible, 84
- sampled, 72
- similarity, 183, 286
- singular, 77, 256
- smoothly varying, 113
- state-output, 77
- time-invariant, 29
- time-reversed, 45
- time-varying, 30
- topological, 123
- underlying, 27
- unstable, 237
 - with no controls, 29
 - with outputs, 27, 45
 - without drift, 158, 393
- tangent vector field, 146
- terminal state, 28
- time
 - series, 77, 312
 - set, 25
- time-invariant, 11
 - i/o behavior, 32
 - system, 29
- time-optimal control, 423
- time-reversed system, 45
- time-varying system, 30
 - stability, 232
- topological system, 123
- tracking, 15, 371
 - error, 372
- trajectory, 28
- transfer
 - function, 333
 - matrix, 333
- transition map, 26, 33
- triple
 - canonical, 284
 - controllable and observable, 283
 - minimal, 286
- UBIBO, 327
- uncontrollable
 - modes, 94
 - polynomial, 94
- unicycle, 164
- uniformly bounded-input
 - bounded-output, 327
- uniqueness of canonical
 - realizations, 286
- universal formula for feedback
 - stabilization, 246
- unstable
 - closed-loop poles, 342
 - open-loop poles, 342
 - system, 237
- value function, 355, 392, 443
- variation of parameters, 488
- vector field, 141
- verification principle, 356
- viscosity solution, 394, 396
- VLSI testing, 310
- Volterra
 - expansion, 75
 - series, 74
- weak
 - controllability, 141
 - convergence, 56, 424
- weakly coercive mapping, 220
- well-posed interconnection, 322
- Young tableaux, 193
- zero
 - measure, 467
 - of a system, 257